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Equidistribution of sequences associated with a-points of the derivatives of the Riemann zeta function

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Abstract. Fujii investigated the uniform distribution of various sequences associated with the non-trivial zeros of the Riemann zeta function by evaluating certain exponential sums over these zeros. In this paper, we present analogous results for a broader class of functions, establishing the uniform distribution of sequences arising from the a-points of the derivatives of the Riemann zeta function.

1. Introduction

1.1. Main theorems. In 1956, Rademacher [11] investigated the uniform distribution properties of the Riemann zeta function. He studied sequences of the form $\{\alpha\gamma\}$, where γ runs over the imaginary parts of the non-trivial zeros of the Riemann zeta function, and obtained uniform distribution results under the assumption of the Riemann Hypothesis. Subsequently, Elliott [5] observed in 1972 that the same result holds unconditionally, and Hlawka [9] later established an unconditional proof. Further generalizations and extensions of these results were obtained in works such as [2,6,10,12], which considered more general frameworks. In particular, Fujii [7] studied sequences of the form $\{f(\gamma)\}$, where $f(x)$ is a function satisfying certain analytic conditions. Examples that can be derived from Fujii's results include the following:

Theorem 1.1 (Fujii [8]). *The sequences like* $\{\gamma\log\gamma/\log_k\gamma\}$, $\{\gamma(\log\gamma)^b\}$, $\{\gamma^{b'}\}$, $\{(\log\gamma)^{b''}\}$, *and* $\{\log\gamma\cdot\log_k\gamma\}$ *are uniformly distributed modulo* 1, *where* k *is a positive integer,* $b<1$, $0<b'\le 1$, *and* $b''>1$.

In this paper, we develop results analogous to those of Fujii. While Fujii's work was restricted to the Riemann zeta function itself, our approach provides theorems that apply to more general situations, including the a-points of the Riemann zeta function and its derivatives. Let c be a sufficiently large constant and let f satisfy the following conditions:

(C1) $f(t)$ is a real-valued C^2 function on (c,∞).
(C2) $tf'(t)^2$ is monotonic on $[c,\infty)$ and remains strictly positive or strictly negative.
(C3) $\dfrac{f'(t)^3}{f''(t)\,\log\bigl(t/(2n_{a,k}\pi)\bigr)}$ is monotonic on $[c,\infty)$ and remains strictly positive or strictly negative (for the precise definition of $n_{a,k}$, see (2.2)).
(C4) $f'(T)^{-1}=o(T)$.
(C5) $\dfrac{f''(T)}{f'(T)^3}=o(T)$.
(C6) $\displaystyle\int_c^T |f'(t)|\log t\,dt=o(T\log T)$.

We can now state our main criterion:

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Theorem 1.2. *If f satisfies the six conditions* (C1)–(C6) *above, then the sequence* $\{f(\gamma_a^{(k)})\}_{\gamma_a^{(k)}>1}$ *is uniformly distributed modulo* 1*, where* $\gamma_a^{(k)}$ *denotes the imaginary part of the* a*-points of the* k*-th derivative of the Riemann zeta function.*

Remark 1.3. As we will see in the proof, we estimate the exponential sum for a-points for the Riemann zeta function and its derivative. If we denote by $N_a^{(k)}(T)$ the number of a-points, the coefficients in the expression for $N_a^{(k)}(T)$ play no essential role in the proof. Thus, the argument applies to any counting function of the form

$$C_1 T(\log T - \log C_2) + O(\log T), \quad C_1 \neq 0,$$

and consequently to a broader class of a-points. In this setting, the function in condition (C3),

$$\frac{f'(t)^3}{f''(t)\,\log(t/(2n_{k,a}\pi))}$$

is replaced by

$$\frac{f'(t)^3}{f''(t)\,\log(et/C_2)}.$$

Under the Riemann Hypothesis, the error term in the zero-counting function $N_0^{(k)}(T)$ of the k-th derivative of the Riemann zeta function admits an improvement, allowing condition (C6) to be replaced by the following weaker condition:

(C6)' $\displaystyle\int_c^T |f'(t)| \frac{\log t}{\sqrt{\log\log t}}\, dt = o(T\log T).$

Theorem 1.4. *Assume the Riemann Hypothesis. If conditions* (C1)–(C5) *hold, together with* (C6)', *then the sequence* $\{f(\gamma_0^{(k)})\}_{\gamma_0^{(k)}>1}$ *is uniformly distributed modulo* 1.

1.2. Examples. The examples presented below are immediate corollaries of our theorems.

Example 1.5. Let $f(t) = ut^v(\log t)^w$ with $u \neq 0$ and $(v,w) \neq (0,0)$. Then

$$f'(t) = ut^{v-1}(\log t)^{w-1}(v\log t + w),$$
$$f''(t) = ut^{v-2}(\log t)^{w-2}\big((2v-1)w\log t + (v-1)v(\log t)^2 + (w-1)w\big).$$

By a simple calculation, for conditions (C4) and (C5) to hold, we must have $v > 0$ or $v = 0, w > 1$. With these conditions in mind, we restrict our attention to these cases. Condition (C6) is satisfied if $v < 1$ or $v = 1, w < 0$ by

$$\int_c^T t^{v-1}(\log t)^w (v\log t + w)\, dt = \begin{cases} O\big(T^v(\log T)^w |v\log T + w|\big) & \text{if } v > 0, \\ O\big((\log T)^{w+1}\big) & \text{if } v = 0, w > 1. \end{cases}$$

Thus, conditions (C4)–(C6) are all satisfied simultaneously in the following cases:

$$0 < v < 1, \quad \text{or} \quad v = 0,\ w > 1, \quad \text{or} \quad v = 1,\ w < 0. \tag{1.1}$$

For sufficiently large c, the monotonicity requirements (C1)–(C3) are satisfied. Hence, the sequence $\{u\gamma^v(\log\gamma)^w\}$ is uniformly distributed modulo 1 when $u \neq 0$ and v and w satisfy (1.1), where $\gamma = \gamma_a^{(k)}$.

Remark 1.6. Fujii's result also covers the borderline case $v = 1, w = 0$ in Example 1.5.

Example 1.7. Assume $v > 1$ or $v = 1, w > 0$. Let $f(t) = u(\log t)^v(\log\log t)^w$ with $u \neq 0$. Then

$$f'(t) = u\frac{(\log t)^{v-1}(\log\log t)^{w-1}(v\log\log t + w)}{t},$$
$$f''(t) = u\frac{(\log t)^{v-2}(\log\log t)^{w-2}}{t^2} \cdot \Big(w(2v - \log t - 1)\log\log t + v(v - \log t - 1)(\log\log t)^2 + (w-1)w\Big).$$

For sufficiently large c, (C1)–(C6) are satisfied, so that the sequence $\{u(\log\gamma)^v(\log\log\gamma)^w\}$ is uniformly distributed modulo 1, where $\gamma = \gamma_a^{(k)}$.

Example 1.8. Assume $0 < v < 1$. Let $f(t) = ut^v\zeta(t)$ with $u \neq 0$. Then

$$f'(t) = uvt^{v-1}\zeta(t) + ut^v\zeta'(t),$$
$$f''(t) = uv(v-1)t^{v-2}\zeta(t) + 2uvt^{v-1}\zeta'(t) + ut^v\zeta''(t).$$

For sufficiently large c, (C1)–(C6) are satisfied, so that the sequence $\{u\gamma^v\zeta(\gamma)\}$ is uniformly distributed modulo 1, where $\gamma = \gamma_a^{(k)}$.

2. Background and previous work

In this paper, we use the following well-known classical results. First, the two lemmas we will use in the proof are as follows:

Lemma 2.1 (cf. Lemma 4.3 of [14]). *Let $F(x)$ and $G(x)$ be real-valued functions, $G(x)/F'(x)$ monotonic, and $F'(x)/G(x) \geq m > 0$, or $\leq -m < 0$. Then*

$$\left|\int_a^b G(x)e^{iF(x)}dx\right| \leq \frac{4}{m}.$$

Lemma 2.2 (Weyl's criterion). *Let $\{x_k\}_{k=1}^{\infty}$ be a sequence of real numbers. Then the sequence $\{x_k\}$ is uniformly distributed modulo 1 if and only if*

$$\lim_{N\to\infty}\frac{1}{N}\sum_{k=1}^{N} e^{2\pi i h x_k} = 0 \quad \textit{for all } h \in \mathbb{Z}\setminus\{0\}.$$

Now let us denote by $N_a^{(k)}(T)$ the number of solutions $\rho_a^{(k)} = \beta_a^{(k)} + i\gamma_a^{(k)}$ to

$$\zeta^{(k)}(s) = a, \quad 1 < \gamma_a^{(k)} \leq T,$$

counted with multiplicity. In connection with Lemma 2.2, we shall employ the classical results concerning $N_a^{(k)}(T)$. The case $k = 0$ corresponds to the ordinary a-points of $\zeta(s)$, while $a = 0$ corresponds to the classical nontrivial zeros. The distribution of the solutions to $\zeta(s) = a$, known as the a-points of the Riemann zeta function, and their analogues for the derivatives $\zeta^{(k)}(s)$, has long been a well-studied topic in analytic number theory.

The following formula, originally established in earlier work, is well-known: for a positive integer k and a non-zero complex number a, one has

$$N_a^{(k)}(T) = \frac{T}{2\pi}\left(\log T - \log(2n_{a,k}\pi e)\right) + A(T). \tag{2.1}$$

Von Mangoldt established the case $k = 0$ and $a = 0$, now known as the Riemann–von Mangoldt formula, with $n_{0,0} = 1$ and error term $A(T) = O(\log T)$. Under the Riemann Hypothesis, this error term improves to $O\Big(\frac{\log T}{\log\log T}\Big)$ (see [14, Theorem 14.13]).

The extension from $a = 0$ to general $a \in \mathbb{C}$ was first obtained by Landau in 1913 [4], in a joint paper with Bohr and Littlewood, with the cited result appearing in the part authored by Landau. He proved that in (2.1) one has $n_{1,0} = 2$ and $n_{a,0} = 1$ if $a \neq 1$. Thus, the Riemann–von Mangoldt formula extends naturally to general a-points.

Berndt [3] obtained the analogue of the Riemann–von Mangoldt formula for $N_0^{(k)}(T)$, that is, for the zeros of $\zeta^{(k)}(s)$ with $a = 0$, proving $n_{0,k} = 2$ $(k \geq 1)$ and $A(T) = O(\log T)$. Assuming the Riemann Hypothesis, this bound improves to $O\Big(\frac{\log T}{\sqrt{\log\log T}}\Big)$ as shown by Akatsuka [1] for the case $k = 1$, and by Suriajaya [13] for the cases $k \geq 2$. The second-named author further established the other cases, $k \geq 1$ and $a \neq 0$, showing that $n_{a,k} = 1$ $(a \neq 0,\ k \geq 1)$ and that $A(T) = O(\log T)$.

Summarizing the above, we obtain

$$n_{a,k} = \begin{cases} 2, & \text{if } (a,k) = (1,0) \text{ or } (a = 0,\ k \geq 1), \\ 1, & \text{otherwise,} \end{cases} \tag{2.2}$$

and the error term satisfies

$$A(T) = \begin{cases} O(\log T), & \text{unconditionally,} \\ O\left(\frac{\log T}{\log\log T}\right), & \text{under the Riemann hypothesis with } a = 0 \text{ and } k = 0, \\ O\left(\frac{\log T}{\sqrt{\log\log T}}\right), & \text{under the Riemann hypothesis with } a = 0 \text{ and } k \geq 1. \end{cases}$$

3. Proofs of Theorems 1.2 and 1.4

Now we prove Theorems 1.2 and 1.4. Differentiating the main term of (2.1) with respect to T, we obtain

$$\frac{d}{dT}\left[\frac{T}{2\pi}\left(\log T - \log(2n_{k,a}\pi e)\right)\right] = \frac{1}{2\pi}\log\frac{T}{2n_{k,a}\pi}.$$

Since $f(t)$ is a real-valued C^2 function on $(c,T]$ with $f'(t) \neq 0$ for all t, by applying Stieltjes integration to the sum over the ordinates $\gamma_a^{(k)}$, we have

$$\begin{aligned} \sum_{c<\gamma_a^{(k)}\leq T} e^{if(\gamma_a^{(k)})} &= \int_c^T e^{if(t)}\, dN_a^{(k)}(t) \\ &= \frac{1}{2\pi}\int_c^T e^{if(t)}\log\frac{t}{2n_{k,a}\pi}\, dt + \int_c^T e^{if(t)}\, dA(t) \\ &=: S_1(T) + S_2(T). \end{aligned} \tag{3.1}$$

For $S_1(T)$, since

$$\frac{d}{dt}\left(\frac{1}{2\pi i}\frac{e^{if(t)}}{f'(t)}\right) = \frac{1}{2\pi}e^{if(t)} - \frac{1}{2\pi i}\frac{e^{if(t)}f''(t)}{f'(t)^2},$$

we have

$$\begin{aligned} S_1(T) &= \int_c^T \frac{d}{dt}\left(\frac{1}{2\pi i}\frac{e^{if(t)}}{f'(t)}\right)\log\frac{t}{2n_{k,a}\pi}\, dt + \frac{1}{2\pi i}\int_c^T \frac{e^{if(t)}f''(t)}{f'(t)^2}\log\frac{t}{2n_{k,a}\pi}\, dt \\ &= \frac{1}{2\pi i}\frac{e^{if(T)}}{f'(T)}\log\frac{T}{2n_{k,a}\pi} + O(1) - \frac{1}{2\pi i}\int_c^T \frac{e^{if(t)}}{tf'(t)}\, dt \\ &\quad + \frac{1}{2\pi i}\int_c^T \frac{e^{if(t)}f''(t)}{f'(t)^2}\log\frac{t}{2n_{k,a}\pi}\, dt. \end{aligned} \tag{3.2}$$

First, we consider applying Lemma 2.1 to

$$\int_c^T \frac{e^{if(t)}}{tf'(t)}\, dt.$$

For this purpose, it suffices that $f(t)$ and $1/(tf'(t))$ are real-valued, and that

$$\frac{1}{tf'(t)^2}$$

is monotonic. Since $tf'(t)^2$ is monotonic and, on $[c,T]$, satisfies

$$tf'(t)^2 \geq \min\{cf'(c)^2, Tf'(T)^2\} > 0.$$

Then Lemma 2.1 yields

$$\int_c^T \frac{e^{if(t)}}{tf'(t)}\, dt \ll \frac{1}{\min\{cf'(c)^2, Tf'(T)^2\}} = O\left(\frac{1}{Tf'(T)^2}\right) + O(1). \tag{3.3}$$

Next, we apply Lemma 2.1 to

$$\int_c^T \frac{e^{if(t)}\, f''(t)}{f'(t)^2} \log \frac{t}{2n_{k,a}\pi}\, dt.$$

In this case, it suffices that $f(t)$ and

$$\frac{f''(t)}{f'(t)^2} \log \frac{t}{2n_{k,a}\pi}$$

are real-valued, and that

$$\frac{f''(t)}{f'(t)^3} \log \frac{t}{2n_{k,a}\pi}$$

is monotonic. By the condition (C3),

$$\frac{f'(t)^3}{f''(t)\log(t/(2n_{k,a}\pi))}$$

is monotonic and, on $[c,T]$, is bounded away from 0 in the sense that

$$\left|\frac{f'(t)^3}{f''(t)\log(t/2n_{k,a}\pi)}\right| \geq \min\left\{\left|\frac{f'(c)^3}{f''(c)\log(c/2n_{k,a}\pi)}\right|, \left|\frac{f'(T)^3}{f''(T)\log(T/2n_{k,a}\pi)}\right|\right\} > 0.$$

Then the required monotonicity holds and Lemma 2.1 gives

$$\begin{aligned}\int_c^T \frac{e^{if(t)}f''(t)}{f'(t)^2} \log \frac{t}{2n_{k,a}\pi}\, dt &\ll \frac{1}{\min\left\{\left|\frac{f'(c)^3}{f''(c)\,\log\frac{c}{2n_{k,a}\pi}}\right|, \left|\frac{f'(T)^3}{f''(T)\,\log\frac{T}{2n_{k,a}\pi}}\right|\right\}} \\ &= O\left(\frac{|f''(T)|\,\log T}{|f'(T)|^3}\right) + O(1).\end{aligned} \tag{3.4}$$

Substituting (3.3) and (3.4) into (3.2), we obtain

$$\begin{aligned}S_1(T) &= \frac{1}{2\pi i}\frac{e^{if(T)}}{f'(T)} \log \frac{T}{2n_{k,a}\pi} + O\left(\frac{1}{Tf'(T)^2}\right) + O\left(\frac{|f''(T)|}{|f'(T)|^3}\log T\right) + O(1) \\ &= O\left(\frac{\log T}{|f'(T)|}\right) + O\left(\frac{1}{Tf'(T)^2}\right) + O\left(\frac{|f''(T)|}{|f'(T)|^3}\log T\right) + O(1).\end{aligned} \tag{3.5}$$

Now we estimate $S_2(T)$. Firstly we have

$$\begin{aligned}S_2(T) &= \left[e^{if(t)}A(t)\right]_c^T - \int_c^T i\, f'(t)\, e^{if(t)} A(t)\, dt \\ &= e^{if(T)}A(T) + O(1) - i\int_c^T f'(t)\, e^{if(t)} A(t)\, dt.\end{aligned}$$

In what follows, we provide distinct proofs for the cases without assuming the Riemann Hypothesis and under its assumption. First, we consider the case without assuming the Riemann Hypothesis. Since $A(T) = O(\log T)$, it follows that

$$S_2(T) = O(\log T) + O\left(\int_c^T |f'(t)|\, \log t\, dt\right). \tag{3.6}$$

Combining (3.5) and (3.6), we obtain

$$\begin{aligned}\sum_{c<\gamma_a^{(k)}\leq T} e^{if(\gamma_a^{(k)})} &= O\left(\frac{\log T}{|f'(T)|}\right) + O\left(\frac{1}{Tf'(T)^2}\right) + O\left(\frac{|f''(T)|}{|f'(T)|^3}\log T\right) \\ &\quad + O(\log T) + O\left(\int_c^T |f'(t)|\, \log t\, dt\right).\end{aligned}$$

From the previous arguments, it remains to show that

$$\frac{\sum_{c<\gamma_a^{(k)}\le T} e^{if(\gamma_a^{(k)})}}{N_a^{(k)}(T)-N_a^{(k)}(c)} \to 0 \quad (T\to\infty).$$

Using the earlier estimates, we have

$$\begin{aligned}\frac{\sum_{c<\gamma_a^{(k)}\le T} e^{if(\gamma_a^{(k)})}}{T\log T} &= O\left(\frac{1}{T|f'(T)|}\right)+O\left(\frac{1}{T^2 f'(T)^2\log T}\right)+O\left(\frac{|f''(T)|}{T|f'(T)|^3}\right)\\ &\quad +O\left(\frac{1}{T}\right)+O\left(\frac{\int_c^T |f'(t)|\log t\,dt}{T\log T}\right).\end{aligned}$$

Therefore we obtain Theorem 1.2, since the above equality converges to 0 under the conditions (C4), (C5), and (C6).

Assuming the Riemann Hypothesis, since $A(T)=\frac{\log T}{\sqrt{\log\log T}}$, and by following the same reasoning as above, we obtain Theorem 1.4.

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